

# Lecture 05: Relationships between two variables

Time plots

Relationships between two  
quantitative variables

Looking at relationships  
visually: Scatterplots

Exploratory analysis using  
scatterplots

Assessing a relationship  
between two variables with  
a number: Pearson's  
correlation

# Recap of previous lectures

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Mostly looking at a single variable:

- ▶ Graphs to explore the distribution of single variables (histograms, bar charts)
- ▶ Summary numbers to describe our distributions:
  - ▶ Measures of central tendency (mean, median)
  - ▶ Measures of spread (standard deviation, IQR)

# Learning objectives for today

- ▶ Time plots to examine what happens to a variable over time
  - ▶ using `geom_point()`
  - ▶ using `geom_line()`
- ▶ Explore the relationship between two quantitative variables
  - ▶ Directionality
  - ▶ Association vs causation
- ▶ Make scatter plots to look at relationships visually
  - ▶ using `geom_point()`
- ▶ Use the correlation coefficient to quantify the strength of linear relationships
  - ▶ calculate correlations using `cor()`

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## Time plots

### Time plots

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# Visualize quantitative variables over time using time plots

- ▶ **Time plots** are a specific subset of plots where the x variable is time.
- ▶ Unlike the previous plots, the time plot shows a relationship between two variables:
  - i) a quantitative variable
  - ii) time
- ▶ Often times, these plots can be used to look for cycles (e.g., seasonal patterns that recur each year) or trends (e.g., overall increases or decreases seen over time).

## Time plots

Relationships between two quantitative variables

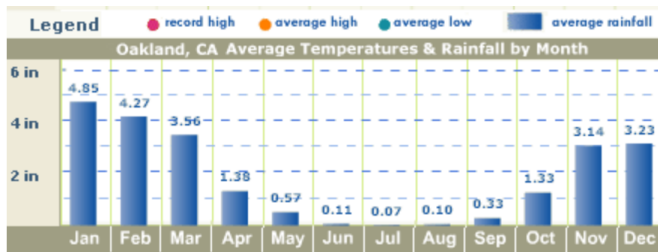
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# Time plot: Oakland rainfall

► from [See California.com](http://SeeCalifornia.com), January 2021:



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# Life expectancy for White men in California

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Make a scatter plot of the life expectancy for White men in California over time.

Since the dataset contains 39 states across two genders and two races, first use a function to subset the data to contain only White men in California.

Which function from last lecture do we need?

► `mutate()`, `select()`, `filter()`, `rename()`, or `arrange()`?

## filter() to select a subset of rows

```
wm_cali <- le_data %>% filter(state == "California",  
                               sex == "Male",  
                               race == "white")
```

*# Equivalent:*

```
wm_cali <- le_data %>%  
  filter(state == "California" &  
         sex == "Male" &  
         race == "white")
```

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## Code for geom\_point()

```
ggplot(data = wm_cali, aes(x = year, y = LE)) +  
  geom_point() +  
  labs(title = "Life expectancy in white men in California, 1969-2013",  
  
        y = "Life expectancy",  
  
        x = "Year",  
  
        caption = "Data from Riddell et al. (2018)")
```

### Time plots

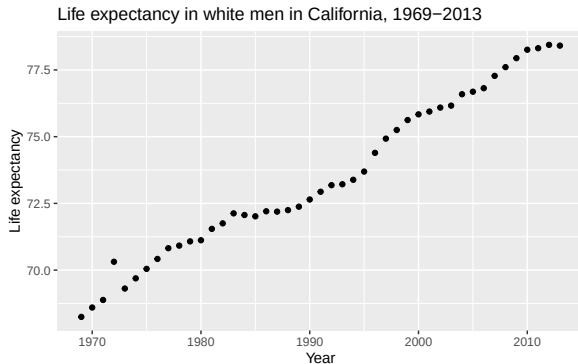
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# Point plot: life expectancy over time



Data from Riddell et al. (2018)

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## Code for geom\_line()

```
ggplot(data = wm_cali, aes(x = year, y = LE)) +  
  geom_line(col = "blue") +  
  labs(title = "Life expectancy in white men in California, 1969-2013",  
  
        y = "Life expectancy",  
  
        x = "Year",  
  
        caption = "Data from Riddell et al. (2018)")
```

### Time plots

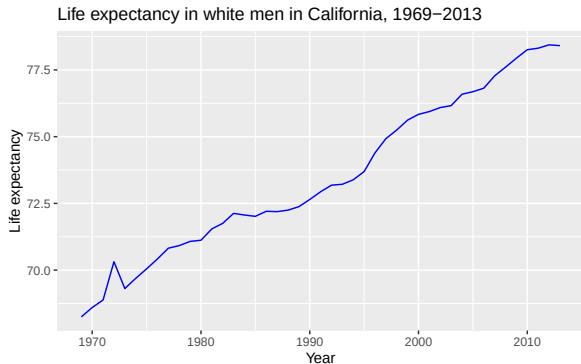
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# Line plot: life expectancy over time



Data from Riddell et al. (2018)

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## Relationships between two quantitative variables

# Explanatory and response variables

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## Bi-directional:

- ▶ “X predicts Y”, or “Y predicts X”
- ▶ “X is associated with Y”, or “Y is associated with X”

## Unidirectional:

- ▶ “X causes Y”

# Which variable is $x$ and which is $y$ ?

In prediction we generally use  $X$  to denote the variable we are using to predict the variable of interest ( $Y$ )

In causation we generally use  $X$  to denote the explanatory (independent) and  $Y$  to denote the response (dependent)

Graphically the  $X$  variable is on the  $X$  (horizontal) axis and the  $Y$  variable is the  $Y$  (vertical) axis

## Practice: choosing $x$ and $y$

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1. Each hospital's rate of hospital-acquired infections, and whether the hospital has implemented a hand-washing intervention as part of a cluster randomized trial.
2. The weight in kilograms and height in centimeters of a person
3. Inches of rain in the growing season and the yield of corn in bushels per day
4. A person's leg length and arm length, in centimeters



# How to investigate causation?

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- ▶ Randomized controlled trials (RCTs) to randomize individuals to different levels
- ▶ Observational study that is *designed* to investigate causation and reduce the risk of bias

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## Looking at relationships visually: Scatterplots

- ▶ Scatterplots are a good way to visualize a relationship between two variables
- ▶ When we look at a scatterplot we want to evaluate:
  - ▶ The overall Pattern of the dots
  - ▶ Any notable exceptions to the pattern
  - ▶ Direction (positive or negative)
  - ▶ Form (straight line or curved)
  - ▶ Strength (how closely the points follow a line)
  - ▶ Are there any obvious outliers

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# Scatterplot syntax in R

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```
name of plot <- ggplot(data = dataset, aes(x = xvariable, y = yvariable)) +  
geom_point(na.rm=TRUE) + theme_minimal(base_size = 15)+  
labs(x = "xlabel", y = "ylabel", title = "Title")
```

# NHANES data

Read in NHANES dataset

```
nhanes_dataNA <- read_csv("nhanes.csv")
nhanes_data <- nhanes_dataNA[rowSums(is.na(nhanes_dataNA[, 15:18])) == 0, ]
nhanes_data %>%
  select(gender, bpxsy1, bpxdi1) %>%
  head()
```

```
## # A tibble: 6 x 3
##   gender bpxsy1 bpxdi1
##   <chr>   <dbl>   <dbl>
## 1 Male     140     90
## 2 Female   136     86
## 3 Female   118     80
## 4 Female   106     60
## 5 Female   122     56
## 6 Male     118     74
```

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# Code: systolic and diastolic blood pressure

```
nhanes_scatter <- ggplot(  
  data = nhanes_data,  
  aes(x = bpxsy1, y = bpxdi1)  
) +  
  geom_point(na.rm = TRUE, size = 0.9, alpha = 0.75) +  
  theme_minimal(base_size = 10) +  
  labs(x = "Systolic BP",  
       y = "Diastolic BP",  
       title = "NHANES Data")
```

Time plots

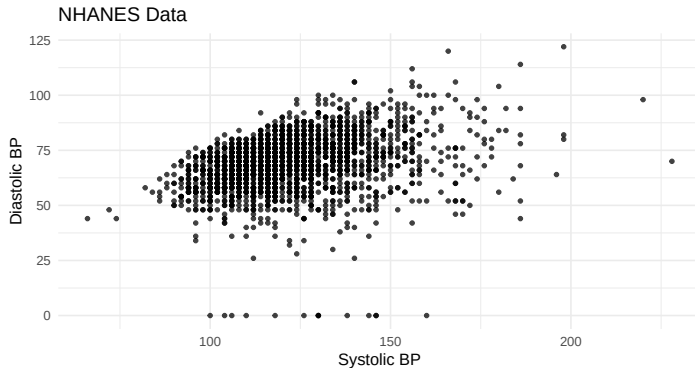
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# Scatterplot: systolic and diastolic blood pressure



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# Discussion: what do we notice?

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What do we notice from the plot?

- ▶ Is there a visible association?
- ▶ Any notable exceptions to the pattern
- ▶ Direction (positive or negative)
- ▶ Form (straight line or curved)
- ▶ Strength (how closely the points follow a line)
- ▶ Are there any obvious outliers



## Adding a third variable with color

We can add a third variable to our graph by coloring the dots

```
nhanes_scatter <- ggplot(  
  data = nhanes_data,  
  aes(x = bpxsy1, y = bpxdi1)  
) +  
  geom_point(  
    aes(col = gender),  
    na.rm = TRUE,  
    size = 0.9,  
    alpha = 0.75  
) +  
  theme_minimal(base_size = 10) +  
  labs(x = "Systolic BP",  
       y = "Diastolic BP",  
       title = "NHANES Data")
```

Time plots

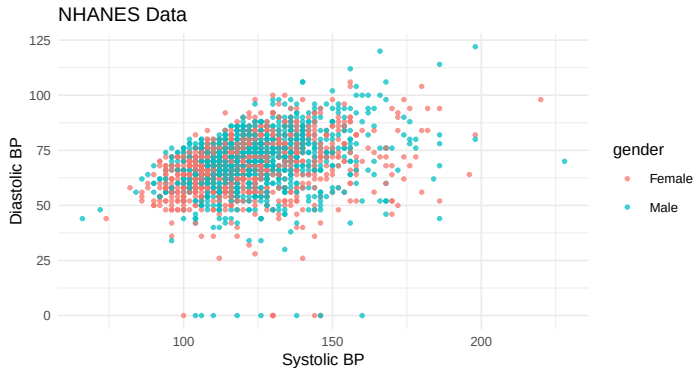
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# Colored scatterplot: blood pressure by gender



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# Manatee data

Manatee data set from your textbook:

```
mana_data <- read_csv("Ch03_Manatee-deaths.csv")  
head(mana_data)
```

```
## # A tibble: 6 x 3  
##   year powerboats deaths  
##   <dbl>      <dbl>   <dbl>  
## 1  1977         447     13  
## 2  1987         645     39  
## 3  1997         755     54  
## 4  2007        1027     73  
## 5  1978         460     21  
## 6  1988         675     43
```

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# Code: powerboats and manatees

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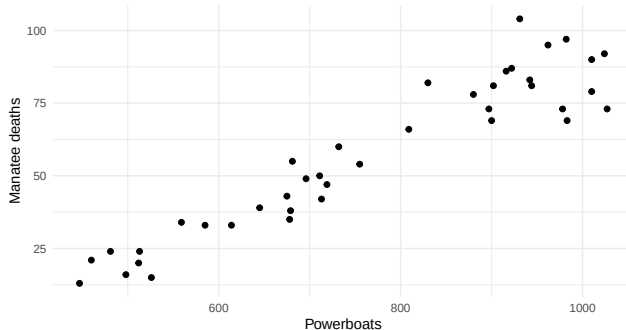
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```
mana_scatter <- ggplot(  
  data = mana_data,  
  aes(x = powerboats, y = deaths)  
) +  
  geom_point(size = 1.5) + theme_minimal(base_size = 10) +  
  labs(x = "Powerboats", y = "Manatee deaths")
```

# Scatterplot: powerboats and manatee deaths



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# Discussion: manatee plot

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# Adding year as a continuous third variable

What if we layer in a continuous third variable?

```
mana_scatter <- ggplot(  
  data = mana_data,  
  aes(x = powerboats, y = deaths)  
) +  
  geom_point(aes(col = year), size = 1.5) +  
  theme_minimal(base_size = 10) +  
  theme(legend.position = "right") +  
  labs(x = "Powerboats", y = "Manatee deaths")
```

Time plots

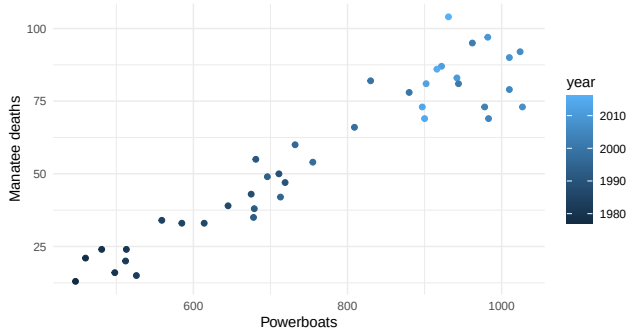
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# Manatee plot colored by year



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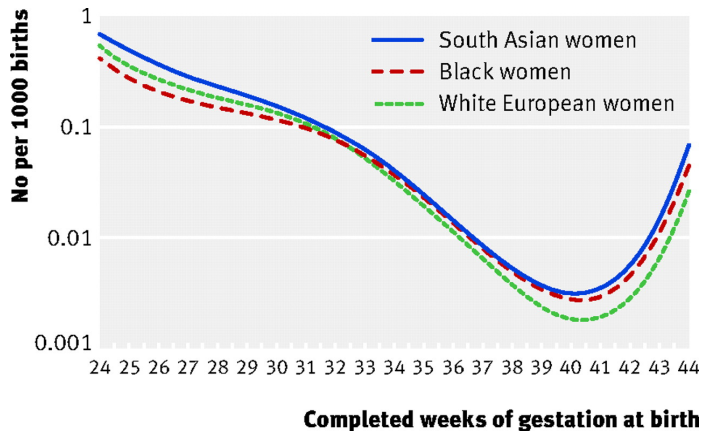
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# A non-linear example: gestational age and perinatal mortality

## Gestational age and perinatal mortality



Source: Balchin et al. BMJ. 2007.

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## Exploratory analysis using scatterplots

# Lean body mass and metabolic rate: Problem and plan

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**Problem:** Is lean body mass (person's weight after removing the fat) associated with metabolic rate (kilocalories burned in 24 hours)?

**Plan:** A diet study was conducted on 12 women and 7 men that measured lean body weight and metabolic rate for each individual.

# Lean body mass and metabolic rate: Data

Data: In the textbook

| Subject | Sex | Mass (kg) | Rate (Cal) | Subject | Sex | Mass (kg) | Rate (Cal) |
|---------|-----|-----------|------------|---------|-----|-----------|------------|
| 1       | M   | 62.0      | 1792       | 11      | F   | 40.3      | 1189       |
| 2       | M   | 62.9      | 1666       | 12      | F   | 33.1      | 913        |
| 3       | F   | 36.1      | 995        | 13      | M   | 51.9      | 1460       |
| 4       | F   | 54.6      | 1425       | 14      | F   | 42.4      | 1124       |
| 5       | F   | 48.5      | 1396       | 15      | F   | 34.5      | 1052       |
| 6       | F   | 42.0      | 1418       | 16      | F   | 51.1      | 1347       |
| 7       | M   | 47.4      | 1362       | 17      | F   | 41.2      | 1204       |
| 8       | F   | 50.6      | 1502       | 18      | M   | 51.9      | 1867       |
| 9       | F   | 42.0      | 1256       | 19      | M   | 46.9      | 1439       |
| 10      | M   | 48.7      | 1614       |         |     |           |            |

What would the corresponding data frame look like? How many variables would it have? How many rows?

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# Creating the weight data frame

Full dataset: 19 rows and 4 variables. Here are the first few rows:

```
head(weight_data, 4)
```

```
## # A tibble: 4 x 4
##   subject gender  mass  rate
##   <dbl> <chr>   <dbl> <dbl>
## 1         1 M         62   1792
## 2         2 M        62.9  1666
## 3         3 F        36.1   995
## 4         4 F        54.6  1425
```

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# Code: scatterplot for weight data

## Exploratory data analysis using scatter plots

```
weight_scatter <- ggplot(  
  weight_data,  
  aes(x = mass, y = rate)  
) +  
  geom_point(size = 1.5) +  
  theme_minimal(base_size = 10)
```

Time plots

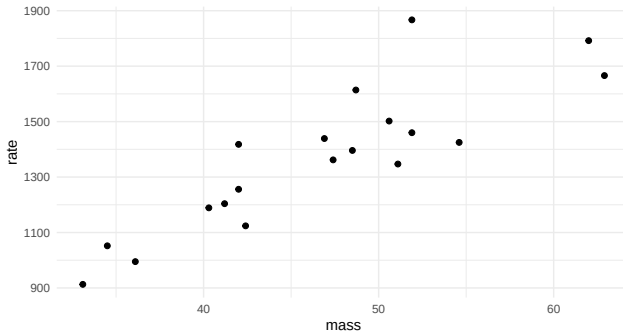
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# Scatterplot: lean body mass and metabolic rate



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## Code: color points by gender

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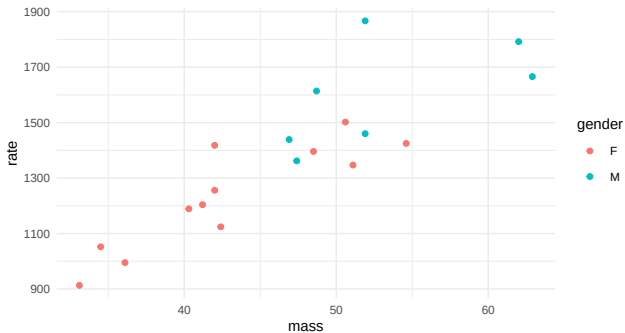
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```
weight_scatter <- ggplot(  
  weight_data,  
  aes(x = mass, y = rate)  
) +  
  geom_point(aes(col = gender), size = 1.5) +  
  theme_minimal(base_size = 10)
```



# Scatterplot colored by gender



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## Code: separate plots for men and women

```
weight_scatter <- ggplot(  
  weight_data,  
  aes(x = mass, y = rate)  
) +  
  geom_point(aes(col = gender), size = 1.5) +  
  theme_minimal(base_size = 10) +  
  theme(legend.position = "none") +  
  facet_wrap(~ gender)
```

Time plots

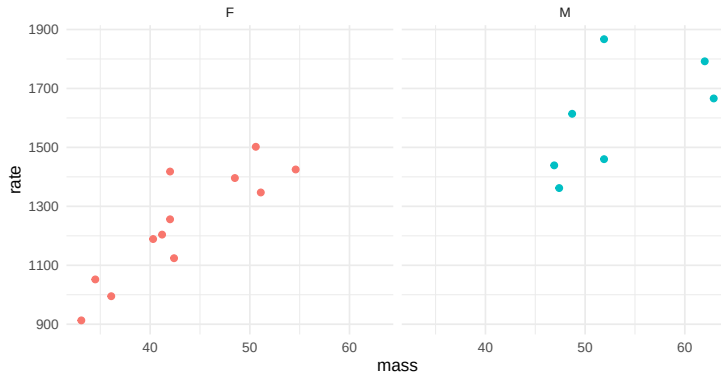
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# Faceted plots by gender



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## Assessing a relationship between two variables with a number: Pearson's correlation

# Pearson's correlation: motivation

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Using just our eyes, we can often say something about whether an association between two variables is weak or strong.

But we can also use a numeric value to describe the direction and strength of an association

# Pearson's correlation formula

- ▶ For linear associations, we can use Pearson's correlation coefficient (denoted by  $r$ ) to quantify the strength of a linear relationship between two variables.
- ▶ The correlation between  $x$  and  $y$  is:

$$r = \frac{1}{n-1} \sum_{i=1}^n \left( \frac{x_i - \bar{x}}{s_x} \right) \left( \frac{y_i - \bar{y}}{s_y} \right)$$

Notice that because we are dividing by the standard deviation the values become unitless

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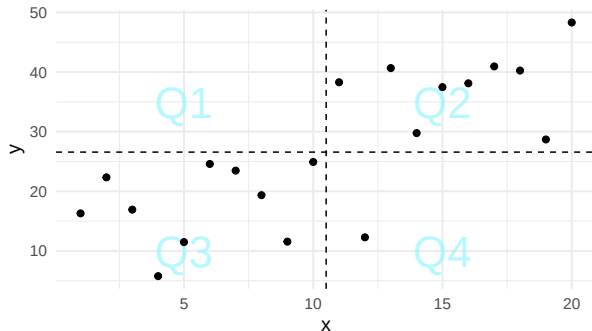
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# Intuition about correlation: center the variables

- ▶ First compare each value to its own mean:  $x_i - \bar{x}$  and  $y_i - \bar{y}$ .
- ▶ The dashed lines split the plot into four regions.
- ▶ Each point is above or below the average value of each variable.



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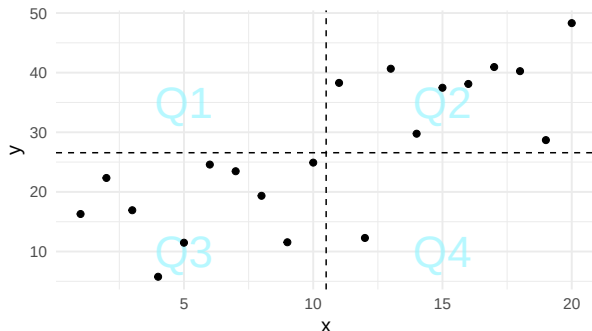
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# Intuition about correlation: signs

$$r = \frac{1}{n-1} \sum_{i=1}^n z_{x,i} z_{y,i}$$

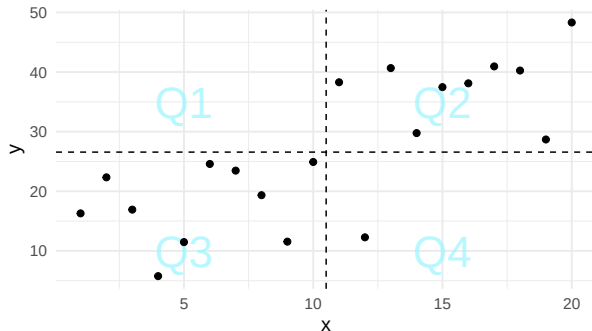
- ▶ Same sign deviations give a **positive** contribution.
- ▶ Opposite sign deviations give a **negative** contribution.





# Intuition about correlation: direction

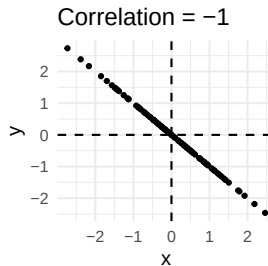
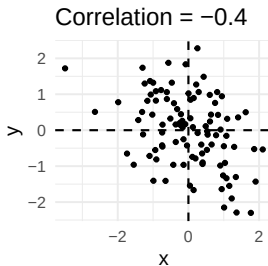
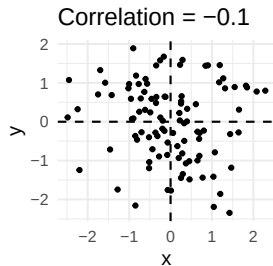
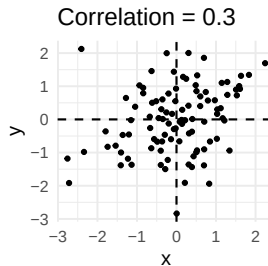
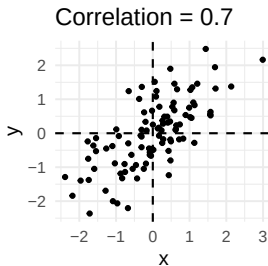
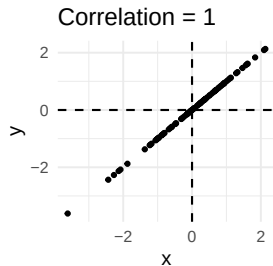
- ▶ More points in Q2 and Q3  $\Rightarrow$  positive correlation.
- ▶ More points in Q1 and Q4  $\Rightarrow$  negative correlation.
- ▶ A balance across regions  $\Rightarrow$  correlation near 0.



# Properties of the correlation coefficient

- ▶ Always a number between -1 and 1.
  - ▶ -1: A perfect, negative linear association
  - ▶ 1: A perfect, positive linear association
  - ▶ 0: No linear association
- ▶ Is used to measure the association between two *quantitative* variables.
- ▶ Only useful for *linear* associations!

# Correlation and direction



Time plots

Relationships between two  
quantitative variables

Looking at relationships  
visually: Scatterplots

Exploratory analysis using  
scatterplots

Assessing a relationship  
between two variables with  
a number: Pearson's  
correlation

# Syntax: Pearson's correlation using `cor()`

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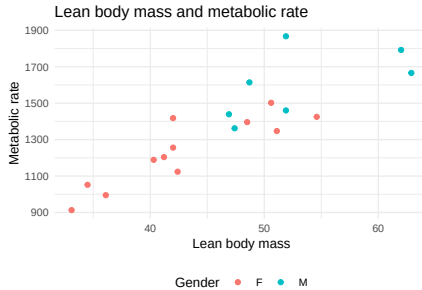
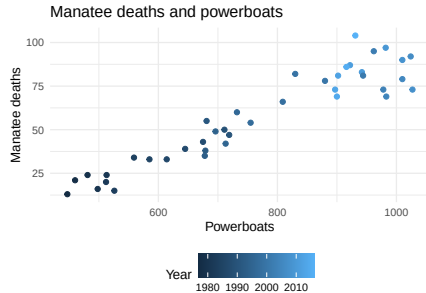
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between two variables with  
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correlation

```
correlation coefficient <- dataset %>%
```

```
summarize(newvar = cor(xvar, yvar))
```

# Remember the manatee and weight plots

Before calculating  $r$ , predict the sign and rough strength from the plots.



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# Correlation for manatee data

Now, calculate the correlations between X and Y for manatees:

```
mana_cor <- mana_data %>%  
  summarize(corr_mana = cor(powerboats, deaths))  
mana_cor
```

```
## # A tibble: 1 x 1  
##   corr_mana  
##   <dbl>  
## 1      0.945
```

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# Correlation for weight data

And for the weight data:

```
weight_cor <- weight_data %>%  
  summarize(corr_weight = cor(mass, rate))  
weight_cor
```

```
## # A tibble: 1 x 1  
##   corr_weight  
##   <dbl>  
## 1      0.865
```

Time plots

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quantitative variables

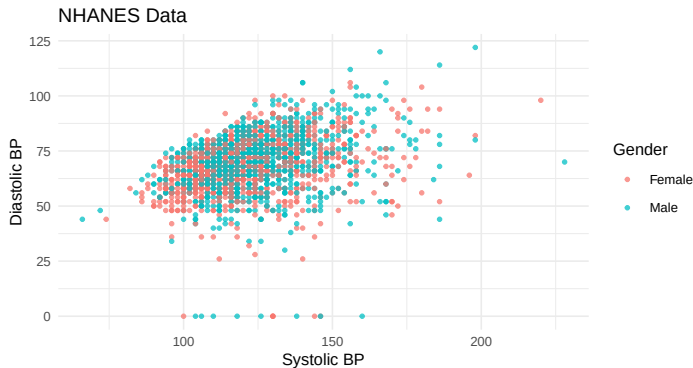
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# Blood pressure plot before correlation

What about our blood pressure data from NHANES?



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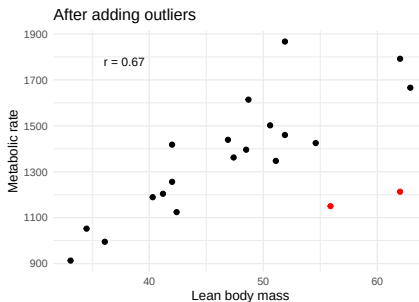
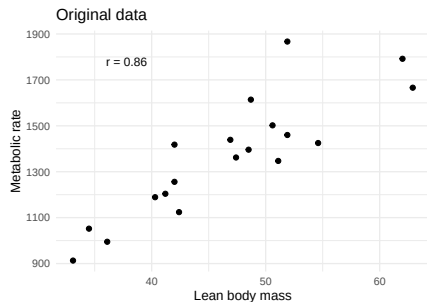
# Correlation for blood pressure data

```
bp_cor <- nhanes_data %>%  
  summarize(corrbp = cor(bpxsy1, bpxdi1))  
bp_cor
```

```
## # A tibble: 1 x 1  
##   corrbp  
##   <dbl>  
## 1  0.322
```

# Correlation is not resistant to outliers

Adding two unusual points changes the correlation for the weight data.



Time plots

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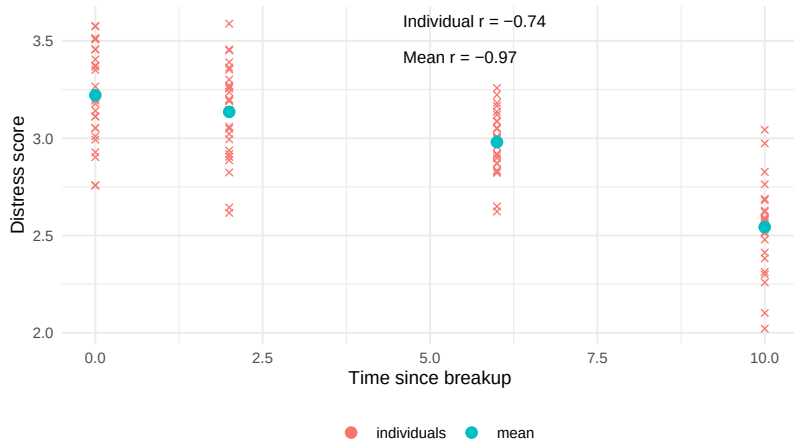
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# Correlations for averages versus individuals

- Correlations based on averages are often stronger than correlations based on individual data.



# Important concepts

- ▶ Determine which variable is explanatory and which is response, or when it doesn't matter
- ▶ Visually describe the relationship between two variables (form, direction, strength, and outliers)
- ▶ Numerically describe the relationship with the correlation coefficient  $r$

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# R Recap: What functions did we use?

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- ▶ `geom_point()`
- ▶ `geom_line()`
- ▶ `aes(col = gender)` to color points by levels of gender
- ▶ `summarize()` to calculate correlation using `cor(var1, var2)`

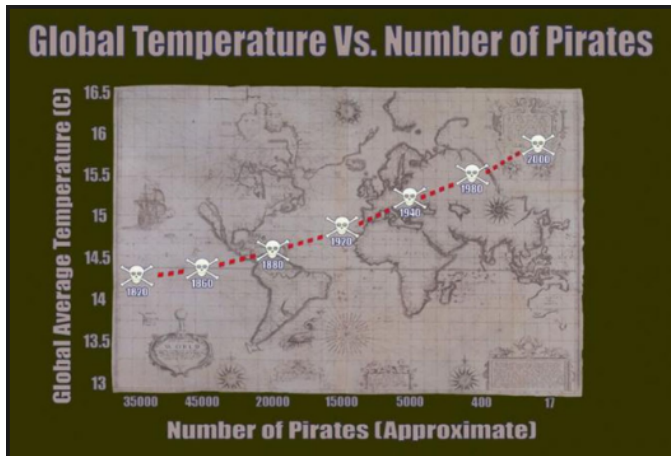
# Reminder: association does not equal causation

Remember that just because two variables are associated, does not mean there is a causal relationship

The correlation coefficient measures association *not* causation.

Even a very strong association doesn't mean that one variable causes the other.

## Reminder: association does not equal causation, pirate example



Example: a strong association can still be non-causal.

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